Web Application Development

IS4102

Lecture Note 04 - **Protocols**

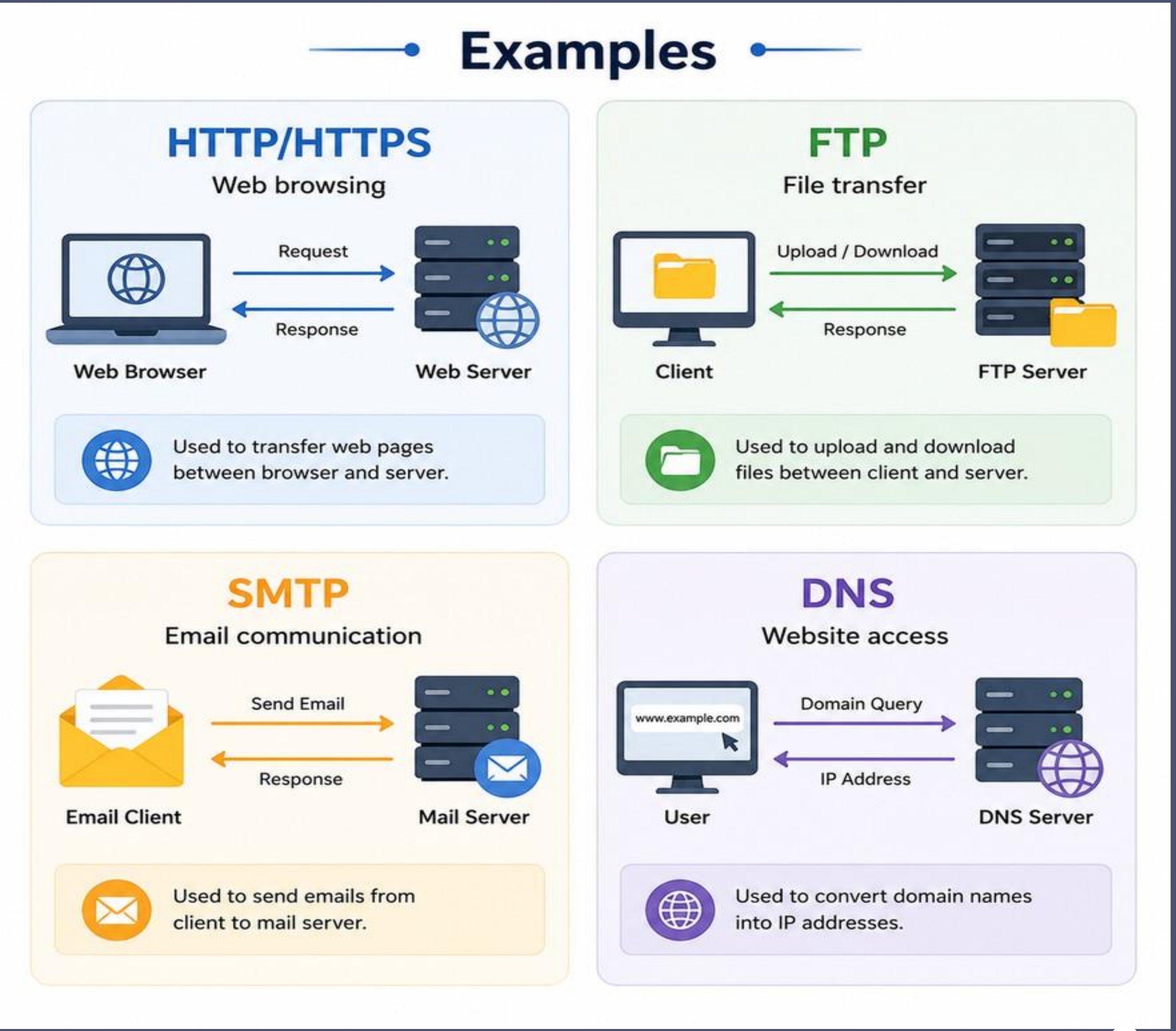
Presented By: Group 21

INTRODUCTION

- A protocol is a set of rules used for communication between devices in a network.
- Protocols ensure that data is transmitted correctly and efficiently.
- They define how data is formatted, sent, received, and processed.
- Common examples include HTTP, FTP, TCP/IP, and SMTP.

WHY DO WE NEED PROTOCOLS?

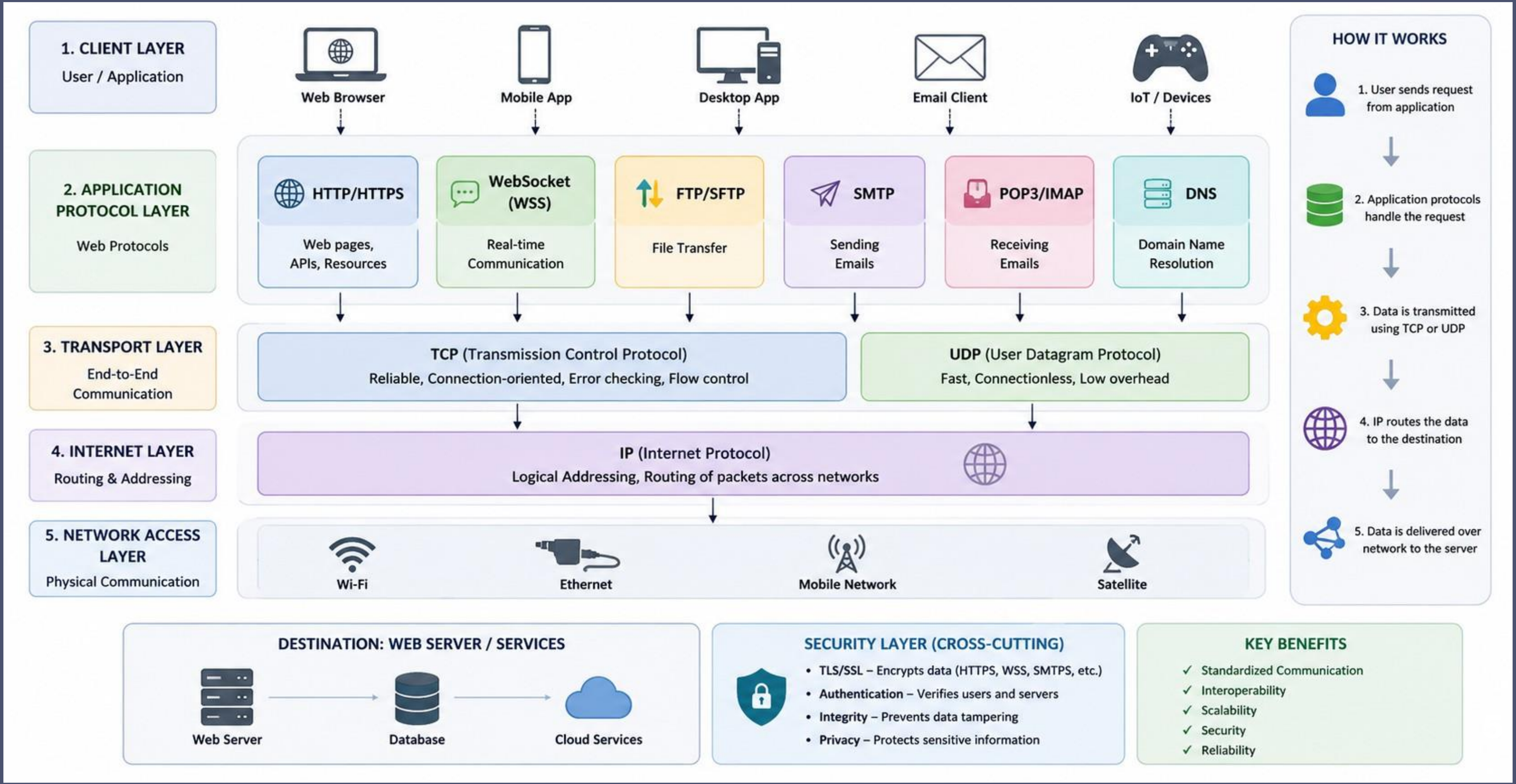
- Enable communication between devices
- Define standard rules for data exchange
- Ensure reliable data transmission
- Improve security and privacy
- Support websites, emails, and file sharing
- Increase network efficiency
- Allow different systems to work together



FEATURES

- Provides reliable communication between devices.
- Supports data transfer over networks.
- Defines rules for error detection and correction.
- Ensures proper data formatting and synchronization.
- Helps maintain secure communication.

ARCHITECTURE



TYPES OF PROTOCOLS

Web/Files	HTTP	HTTPS	DNS
	TPC/IP	FTP	
Network Protocols	IP	DHCP	ARP
Security Protocols	SSL/TLS	SSH	

Activity

Find out about the different Protocol Types

ADVANTAGES

- Enables fast and reliable communication.
- Improves network efficiency. Supports communication between different devices and systems.
- Provides better security and error handling.
- Helps manage large amounts of data transmission.

DISADVANTAGES

- Some protocols are complex to configure.
- Security risks may occur if protocols are not properly protected.
- Network congestion can reduce performance.
- Certain protocols require high bandwidth and resources.

REAL-WORLD APPLICATIONS

- Web browsing and internet communication.
- Email services and file sharing.
- Online banking and e-commerce systems.
- Video conferencing and social media platforms.
- Cloud computing and IoT devices.

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